

Zach Martim

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EDUCATION

University of Utah – Kahlert School of Computing

Bachelor of Science, Computer Science – AI/ML and Information Tracks; GPA: 3.6

Salt Lake City, UT

Aug 2022 – Dec 2026

- **Relevant Coursework:** (Adv.) Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Algorithms, Database Systems, Computer Systems, Digital Image Processing, Foundations of Data Analysis

Brigham Young University

Bachelor of Science, Computer Science (transferred to the University of Utah)

Provo, UT

Aug 2021 – Apr 2022

EXPERIENCE

Cimento AI

Software Engineer Intern

Salt Lake City, UT

Jan 2026 – Present

- Built and shipped a standalone internal executive analytics service in Next.js and TypeScript, delivering 15 metrics across 4 views with 7-, 30-, and 90-day filtering over a multi-tenant Aurora PostgreSQL database.
- Designed a schema-scoped Kysely client serving every tenant schema from one shared connection pool rather than a pool per tenant, avoiding mutable search-path state, and parallelized queries so a failing tenant could not break company-wide reporting.
- Provisioned the service's AWS infrastructure across 23 Terraform resources (ECS, ECR, ALB, IAM, Secrets Manager, RDS Proxy) with a Dockerized GitHub Actions CI/CD pipeline and a read-only database role.
- Replaced a lossy rich-text editor with a dual visual/source phishing template editor (contenteditable iframe, CodeMirror 6) preserving customer-authored email HTML, adding HTML/EML import and 2 independent stored-XSS sanitization gates (DOMPurify at storage, Go blumonday at send).
- Built a production MCP server from the ground up as an OAuth 2.1 resource server (Supabase, Okta SAML, PKCE) exposing 8 read-only tools, with audience-bound JWTs verified against JWKS, tenant-admin authorization, and tenant-scoped audit logging.
- Partnered with a design-partner customer through weekly calls and live demos, translating their requirements into shipped features and validating releases against their feedback.

ServiceNow, University of Utah

Software Engineer Intern

Salt Lake City, UT

Dec 2022 – Jan 2026

- Engineered custom workflows, forms, KPIs, reports, and modules within ServiceNow, enhancing system functionality and user experience while working on a team of 15+ developers.
- Resolved 150+ system bugs within the University of Utah's IT infrastructure, ensuring uninterrupted functionality and improving system uptime.
- Designed and implemented a ServiceNow integration with Cisco ISE via the ERS REST API, routing authenticated requests through MID Servers to manage network endpoints.

PROJECTS

Lumen | React, TypeScript, Firebase, Vertex AI | Senior Capstone | Team Lead (3 engineers)

Jan 2026 – Present

- Built an LLM authoring harness generating a complete medical simulation scenario (patient persona, assessment rubric, conditional feedback rules) through a 3-stage conversational workflow, replacing a 45-field manual form.
- Architected a plugin-based OSCE case system with 9 slide types behind a discriminated-union config and editor dispatcher, spanning authoring, student runtime, and a transactional publish pipeline with immutable version snapshots.
- Built the real-time speech-to-speech audio engine: AudioWorklet PCM16 capture at 16 kHz, gapless 24 kHz playback, and a dual-channel WAV encoder reconciling bursty WebSocket audio into a correctly timed stereo recording for graders.
- Partnered with 2 medical students across the full build, refining the authoring workflow and case format against their feedback and observed practice sessions with other medical students.

HACKATHON AWARDS

TicketMaster Hack the SDLC | 1st Place

Dec 2025

- Created an AI system converting meeting recordings into structured, ready-to-import Jira tickets that users can edit or adopt directly.

Rover-Q | University of Utah x Redo Hackathon | 1st Place

Oct 2025

- Engineered a frontend experience where users can create a best path for a Mars Rover using Q-learning reinforcement learning.

Pre-Launch Product Analytics | Redo Agentic AI Hackathon | 3rd Place

Aug 2025

- Built an agentic AI platform modeling user behavior through profile-driven agents, enabling companies to forecast product performance and optimize e-commerce strategies.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, C#, Java, Go, SQL

Frameworks: React, Next.js, Node.js, FastAPI, Firebase Cloud Functions, .NET/MAUI, Qt

Cloud/Infrastructure: AWS (ECS, ECR, ALB, RDS, S3, DynamoDB, AppSync), Terraform, Docker, GitHub Actions, Linux

Databases/Tools: PostgreSQL, Firestore, GraphQL, Supabase, ServiceNow, Git, Postman, Jupyter Notebook